

Deep Learning for Text Analytics

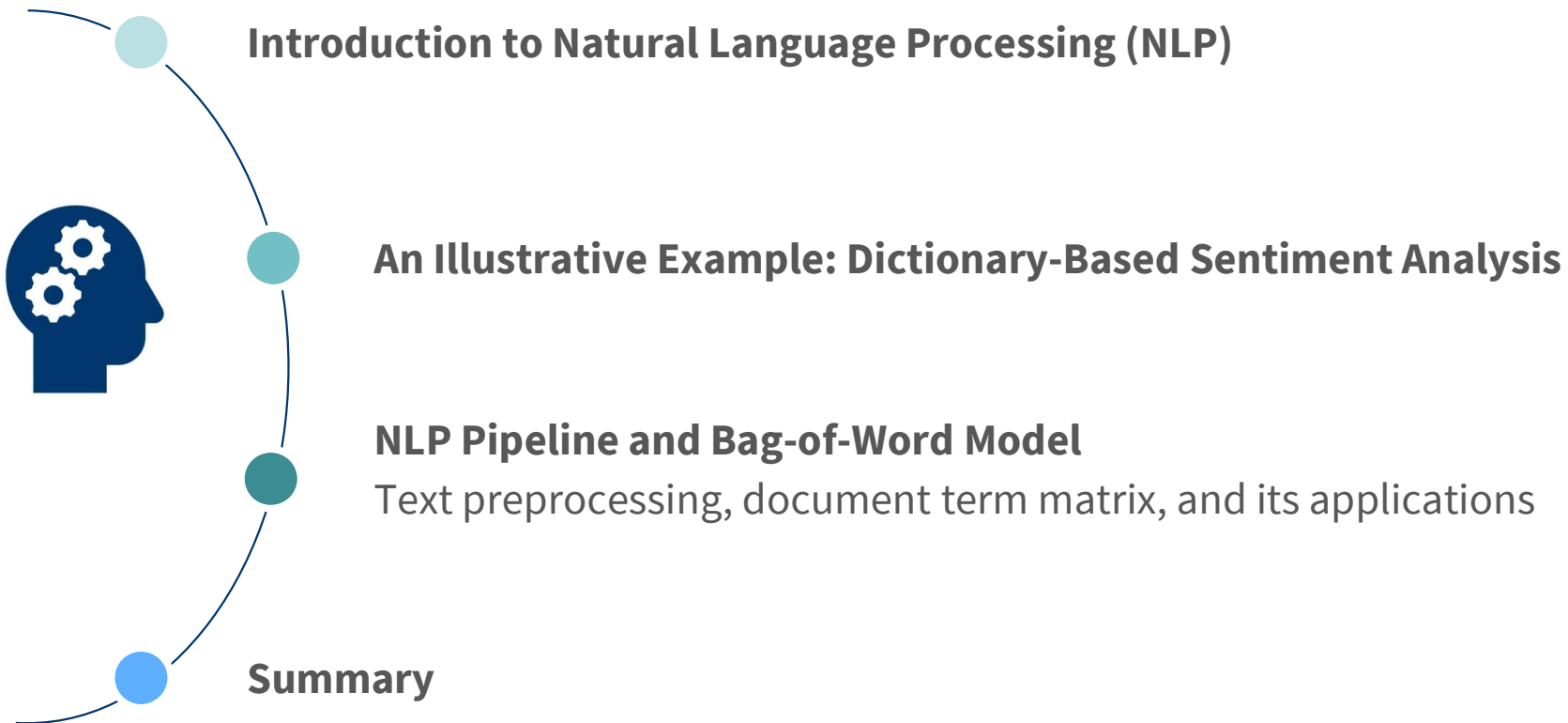


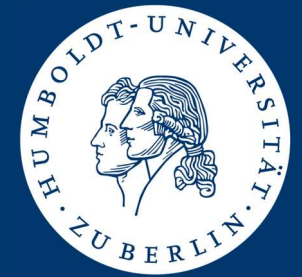
Foundations of Natural Language Processing

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Agenda

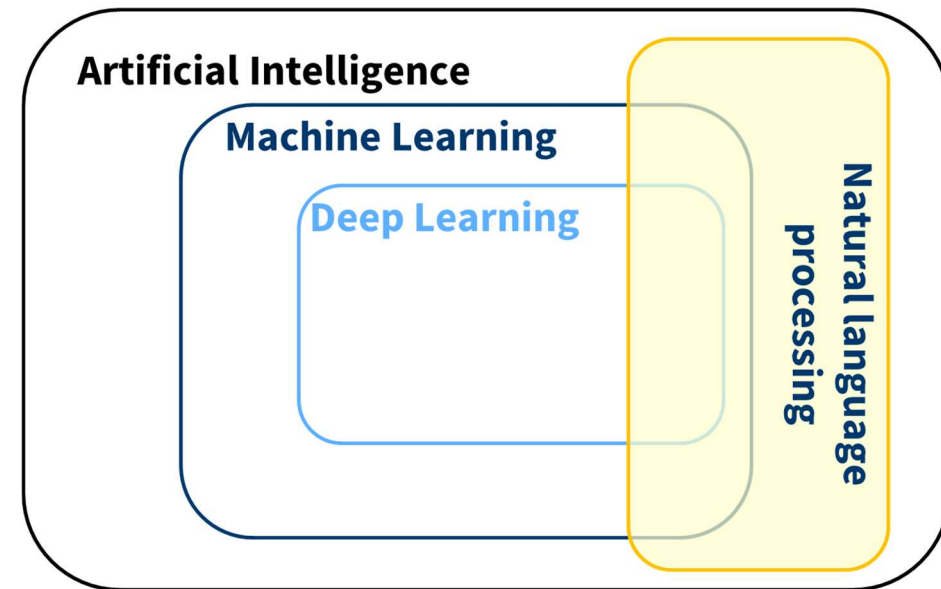




Introduction to Natural Language Processing (NLP)

Natural Language Processing (NLP)

A set of linguistic, statistical & ML techniques that enable computers to process, analyze, and ultimately understand textual data including contextual nuances and capturing the informational content within such data.



Why Natural Language Processing?

Unlock informational value of text data & build smart (AI) systems

■ Explosion of unstructured text data

- Most of the world's data is unstructured, and a huge portion is text
 - Communication with customers, business reports, medical records, newspaper articles
 - No to mention social media
- NLP is how we can unlock the informational value in such data
 - Unstructured data is not readily available in transactional databases
 - Unstructured data is not readily usable in descriptive/explanatory/predictive models

■ AI effect - language is the interface of intelligence

- Human language is the primary vehicle for thought, knowledge, and interaction
- Enabling machines to understand and generate language is key to artificial intelligence
 - Daily experiences powered by NLP: voice assistants, search engines, autocomplete, translation tools, ...
 - Multimodal foundation models are at the core of state-of-art AI platforms

Evolution of Natural Language Processing

NLP has evolved through distinct eras, each solving key challenges

- **Early paradigm:** *if language is governed by rules, let's write those rules*
- **Next idea:** *let the data tell us how language works*
- **Current idea:** *learn the text representation and how to solve tasks end-to-end*

Era	Approach	Key Idea	Limitations
Symbolic (1950s – 1980s)	Rules/Logic	Encode linguistic knowledge	Inflexible, error prone, labor-intensive
Statistical (1990s – 2010s)	Probabilistic	Learn from labeled corpora	Scarcity of labels, Sparse, shallow semantics
Deep Learning (2013 –)	Neural Networks	Learn end-to-end from data	Black-box, data-hungry

Evolution of Natural Language Processing

Perspectives, inputs, and outputs have continuously evolved

Model Type	Input Unit	Output Unit Represents	Dimensionality	Notes
Bag of words (e.g., TF-IDF)	Document	Document	High / sparse	No order or semantics
Word Embeddings (e.g., Word2Vec)	Word	Word	Low / dense	Captures semantics
Transformers (e.g., GPT)	(short) piece of text	(short) piece of text	Low / dense	Captures semantics and context
Multimodal LLMs	Text, image, audio, etc.	Text, image, audio, etc.	Higher	Semantics and context across data types
???	???	???	???	???

An Illustrative Example: NLP for Feature Extraction

Extract information from text to create features for analytical models

■ Augment ML models with auxiliary features derived from text data

- Continued popularity in research
- Business research examples
 - Consumer behavior: textual data can shed light on a variety of behavioral and attitudinal traits
 - Innovation management: patent data as cue on technological progress
 - ESG/CSR reports can shed light on companies' approach toward sustainability
 - Social media discourse can inform about investor sentiment, market efficiency, etc.

■ Somewhat old-fashioned?

- Unstructured text data treated as static source of information
- Consider this an ML approach, as opposed to an AI approach



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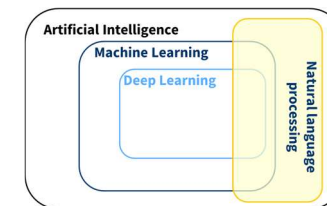
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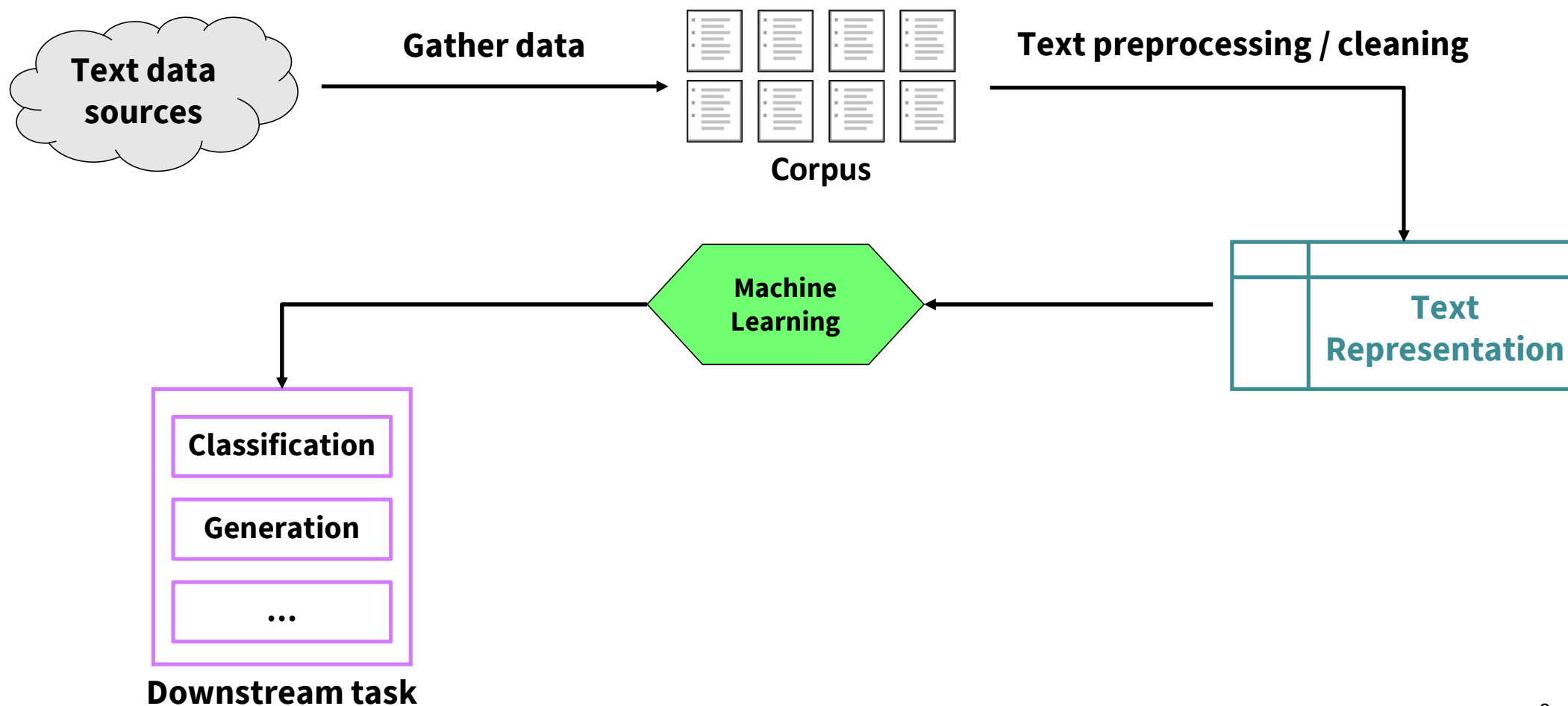
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An Illustrative Example: NLP for Feature Extraction

NLP pipeline maps text to a numerical representation as ML model input



Common NLP Task and Applications

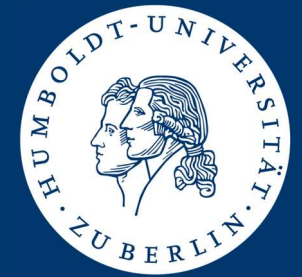
Upstream tasks are building blocks to address downstream tasks

Upstream tasks

- **Language identification**
- **Tokenization**
 - Splitting text into units
 - Words, subwords, sentences, ...
- **Part-of-speech (POS) tagging**
 - Identify grammatical role of a piece of text
 - Noun, verb, adjective, adverb, etc.
- **Named Entity recognition (NER)**
 - Locate and classify named entities
 - “*The price of Apple rose yesterday*”
- ...

Downstream tasks

- **Sentiment analysis**
- **Text classification & clustering**
- **Text summarization**
- **Topic modeling**
- **Machine translation**
- **Question answering**
- **Text generation**
- ...



An Illustrative Example: Dictionary-Based Sentiment Analysis

Simple NLP Example: Dictionary-Based Sentiment Analysis

Lookup word-level polarity scores in a dictionary and aggregate

■ Goal is to predict the polarity of a piece of text

■ Sentiment dictionary

- Pre-defined set of opinion words (mainly adjectives)
- Intensity scores or binary sentiment per word
- Specific dictionaries for domain-specific words
 - bullish, bearish, moon, rally, ATH, ...
 - Famous example: [LM Dictionary](#) for finance/accounting

■ Process

- Process each word in a text individually
- Check whether word appears in the lexicon
- If yes, retrieve corresponding sentiment score
 - Detect negators (not, never, ...)
 - Adjust for intensifiers/diminishers (very good, rather bad, ...)
- Aggregate scores over all words

Word	Sentiment	Score	Notes
love	Positive	+0.9	Strongly positive emotion
like	Positive	+0.5	Mildly positive
happy	Positive	+0.8	Emotional state
amazing	Positive	+0.95	Superlative
okay	Neutral	0.0	No strong sentiment
boring	Negative	-0.6	Low engagement
bad	Negative	-0.7	General negative
terrible	Negative	-0.95	Strong negative
angry	Negative	-0.9	Emotion-based negative
not	Negator	—	Used for negation (not happy, etc.)

Exemplary sentiment lexicon

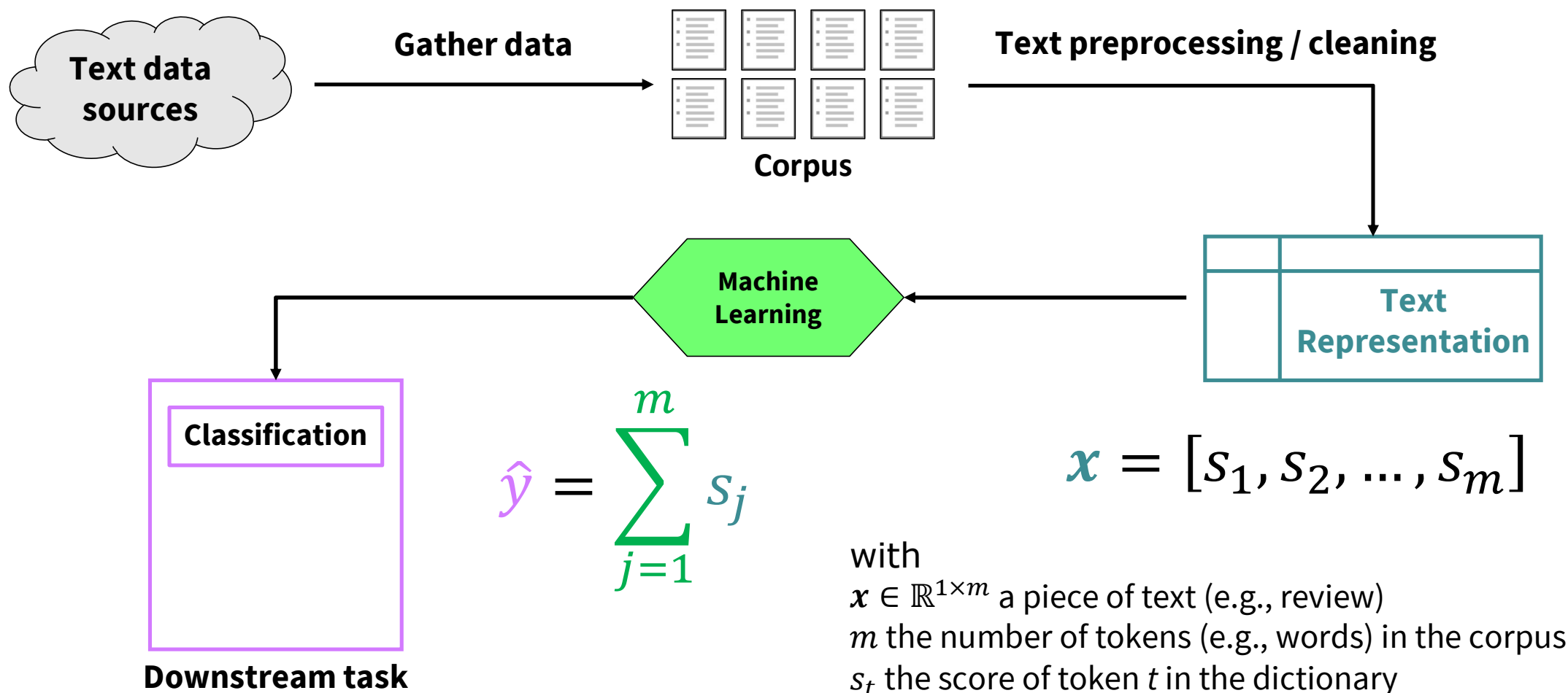
Strengths and Weaknesses of Dictionary-Based Sentiment Analysis?

Strengths and Weaknesses of Dictionary-Based Sentiment Analysis?

- **Simple and interpretable**
- **No training data needed**
- **Fast and lightweight**
- **Domain-adaptable**
- **Can be combined with rules**
 - Negations
 - Intensifiers
 - Diminishers
- **Ignores context**
- **Fixed, static vocabulary**
- **Negations and modifiers are tricky**
 - For example, sarcasm, irony, ...
 - „Oh great, another Monday morning“
- **Does not learn from data**
- **Low coverage of domain-specific jargon**

Dictionary-Based Sentiment Analysis in the NLP Pipeline

Represent tokens by their (weighted) polarity score. Summed scores as prediction



Dictionaries Remain Popular and Exists for Several Applications

NLP process remains the same as in sentiment analysis

■ Emotion and affect lexica

- Emotional states (e.g., joy, anger, fear, panic)
- Psychological categories (e.g., certainty, inhibition)
- Personality traits (e.g., OCEAN model, aka big five)

■ Concreteness ratings

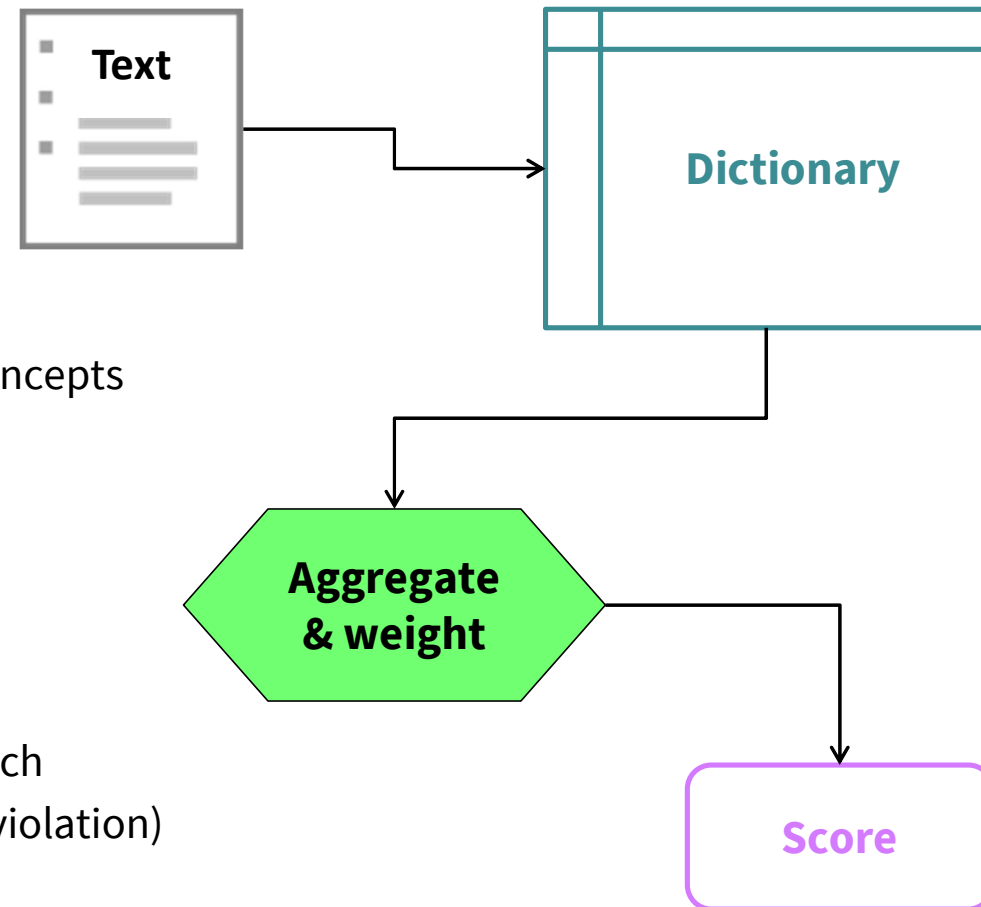
- Scales to discriminate between abstract and concrete concepts
- For example: "justice" → abstract, "car" → concrete

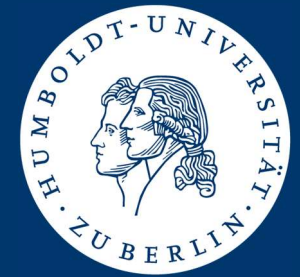
■ Lexical semantics and word similarity

- Word pair similarity scores (e.g., "car" vs "vehicle")
- Word association strengths (e.g., "doctor", "hospital")

■ Toxicity, bias, social norms

- Hate speech, threat, insult, and other forms of toxic speech
- Social biases and moral norms (e.g., "cheat" → fairness violation)





NLP Pipeline and Bag-of-Word Model

Text preprocessing, document term matrix, and its applications

NLP Preprocessing Pipeline

Step 1: Tokenization

The lecture ended on time.

■ Split text as a stream of characters into individual tokens

- Can be done at word level (e.g., using whitespaces)
- Typically done at sub-word level today
 - “Boyfriend” → “boy”, “friend”
 - More flexible and less tokens
 - Can even consider character level



Token	Token	Token	Token	Token	Token
The	Lecture	ended	on	time	.

■ Results in a bag of words/tokens

NLP Preprocessing Pipeline

Step 2: Filtering

The lecture ended on time.

■ Filters for **stop word, punctuations, numbers and symbols**

- Often considered irrelevant for downstream tasks
- Reduce number of distinct tokens

■ Based on dictionaries

- Specific to individual languages
- Risk to change semantics of a piece of text

Token	Token	Token	Token	Token	Token
The	Lecture	ended	on	time	.

Token	Token	Token	Token	Token	Token
The	Lecture	ended	on	Time	.

Document Term Matrix (DTM)

Early approach representing text data in a tabular format

■ Corpus

- Collection of documents
- E-mails, news articles, tweets, etc.

■ Vocabulary

- Set of unique words in the corpus
- After NLP preprocessing

■ Documents are represented – **embedded** – as numerical vectors

- Similar to structured data
- Feature values characterize observations

■ DTM characteristics

- High dimensionality (size of vocabulary)
- Sparsity (words appear in only a few documents)

Other Steps in the NLP-Pipeline: Stemming/Lemmatization

Map words to their base form/grammatical root

■ Goal is to reduce the sizes of the vocabulary

- Less distinct tokens
- Less columns in the DTM

■ Reducing a word to its most basic element (*stem*)

■ Determining the dictionary form (*lemma*) of word

- *run, runs, ran, running* → *run*
- *am, are, is* → *be*

■ Lemmatization typically superior & more costly

- No variance due to stemmer choice
- Dictionaries and morphological analysis needed
- Dictionaries are specific to a language

Token	Token	Token
Lecture e	ended ed	Time

Token	Token	Token
Lectur	end	Time

	Standard stemmers		
	Porter	Snowball	Lancaster
Generate	Gener	Generat	Gen
Generates	Gener	Generat	Gen
Generated	Gener	Generat	Gen
Generating	Gener	Generat	Gen
General	Gener	General	Gen
Generally	Gener	General	Gen
Generic	Gener	Generic	Gen
Generically	Gener	Generic	Gen
Generous	Gener	Generous	Gen
Generously	Gener	Generous	Gen
Organization	Organ	organ	Org
urgency	urgen	urgen	Urg

Other Steps in the NLP-Pipeline: Speech (POS) Tagging

■ Identify the grammatical role of a word

- Noun, verb, adjective, etc.
- Based on dictionaries

■ Standard (up-stream) NLP task

- Typically supports other NLP tasks
- Example is lemmatization

■ Intuition:

- Given dictionary with known nouns, verbs, etc.
- Go through corpus
 - Calc frequency of nouns following 'to' $P(\text{Noun}|\text{To}) = 0.001$
 - Calc frequency of verbs following to' $P(\text{Verb}|\text{To}) = 0.83$
 - Predict more probable grammatical role

NO.	TAG	DRSCRIPTION
1.	CC	Coordinating conjunction
7.	JJ	Adjective
8.	JJR	Adjective, comparative
13.	NNS	Noun, plural
14.	NNP	Proper noun, singular
15.	NNPS	Proper noun, plural
19.	PRP\$	Possessive pronoun
20.	RB	Adverb
21.	RBR	Adverb, comparative
24.	SYM	Symbol
25.	TO	to
27.	VB	Verb, base form
29.	VBG	Verb, gerund or present participle
30.	VBN	Verb, past participle
32.	VBZ	Verb, 3rd person singular present

Dictionary example: <https://catalog.ldc.upenn.edu/docs/LDC95T7/cl93.html>

NNP	VBZ	VBN	TO	VB	RB
Ronaldo	is	expected	to	play	tomorrow
NNP	VBZ	VBZ	TO	NN	RB
Ronaldo	is	expected	to	play	tomorrow

Document Term Matrix (DTM) Applications and Alternatives

Binary, count-based, and weighted document embeddings for standard ML

■ DTM facilitates **document-level** analysis

- Standard ML methods readily applicable
- Regression, classification, clustering, etc.

■ Common document representations

- Binary embeddings measure whether a token occurs in a document
- Count-based embeddings measure how often a token occurs in a document
- Weighted counts to capture a word's relevance for a document
 - Words that occur often in the corpus do not tell much about a specific document
 - Term-frequency-inverse-document-frequency (TFxIDF) embeddings consider the ration of document-level versus corpus level word frequency

← Vocabulary →

ID	DESPITE	WORST	ENJOY	MOVIE	COOL
1	1	0	1	0	1
2	0	0	0	1	0
3	0	1	0	1	0
4	0	1	1	1	1

ID	DESPITE	WORST	ENJOY	MOVIE	COOL
1	1	0	2	0	2
2	0	0	0	2	0
3	0	3	0	6	0
4	0	1	2	8	2

Term Frequency Inverse Document Frequency (TFxIDF)

Assign high weight to terms that are frequent in a document but rare across the corpus

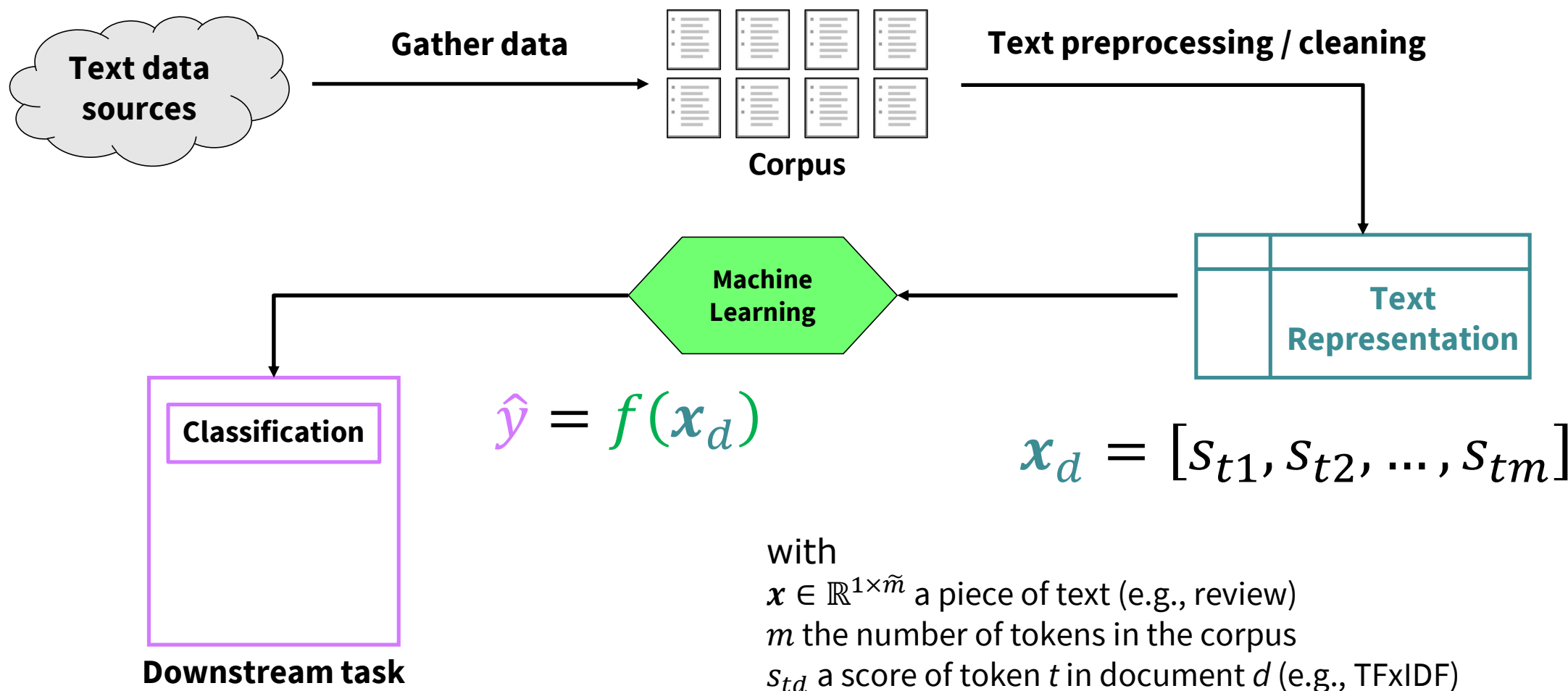
$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log\left(\frac{N}{\text{DF}(t)}\right)$$

with

- **t : a term (word or token)**
- **d : a document**
- **TF(t, d): term frequency**
 - how often term t appears in document d
 - Raw count or normalized frequency
- **DF(t): document frequency**
 - number of documents in the corpus that contain term t
- **N : total number of documents in the corpus**

Dictionary-Based Sentiment Analysis in the NLP Pipeline

Represent tokens by their (weighted) polarity score. Summed scores as prediction



Quo Vadis Traditional NLP?

Strengths and weaknesses of traditional NLP and the DTM

■ Strengths

- ☐ ...
- ☐ ...
- ☐ ...

■ Weaknesses

- ☐ ...
- ☐ ...
- ☐ ...

DOC ID	TOKEN ID					
	1	2	3	4	5	...
1	0	1	0	1	1	...
2	0	0	1	0	0	...
3	1	0	1	0	1	...
4	1	1	1	1	0	...
...	...	...	...	...	...	...

Shortcomings of DTM and Traditional NLP

No understanding of the rich sources of information in language

■ Bag of words model ignores word order

- Contextual information is lost
- “This is interesting” vs. “Is this interesting”

■ Problems with synonyms

- Vocabulary inflation as words with similar meaning map to different tokens
- Risk of semantic segmentation
 - “The students were happy” vs. “The students were joyful”
 - Numerically, these sentences look more different than they are
- Poor generalization if a synonym (e.g., joyful) was not in the corpus

■ Problems with homonyms

- Words with same/similar spelling map to the same token
- Loss of semantic information
- “She went to the bank to withdraw money.” vs. “She sat on the bank of the river.”

■ Technical challenges related to the DTM

DOC ID	TOKEN ID					
	1	2	3	4	5	...
1	0	1	0	1	1	...
2	0	0	1	0	0	...
3	1	0	1	0	1	...
4	1	1	1	1	0	...
...	...	...	...	...	...	...



Summary

Summary



Learning goals

- NLP foundations
- BoW model and DTM



Findings

- Dictionaries are a popular, simple form of NLP for text classification
- NLP preprocessing steps allow us to represent text data as numerical vectors
- BoW model ignores word order, semantics, and context
- DTM facilitates document-level machine learning but suffers from high dimensionality and sparsity



What next

- Lecture: Reuseable dense word embeddings
- Tutorial: NLP foundations

Thank you for your attention!

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